

Section A

1. (a) (i) $a = 0.805084\dots$ and $b = 2.88135\dots$
 $a = 0.805$ and $b = 2.88$ **A1A1**
- (ii) $r = 0.97777\dots$
 $r = 0.978$ **A1**
- [3 marks]**
- (b) a represents the (average) increase in waiting time (0.805 mins) per additional customer (waiting to receive their coffee) **R1**
- [1 mark]**
- (c) attempt to substitute $x = 7$ into their equation **(M1)**
- 8.51693...
- 8.52 (mins) **A1**
- [2 marks]**
- Total [6 marks]**

2. (a) attempt to use sine rule (M1)

$$\frac{24}{\sin 113^\circ} = \frac{17}{\sin \hat{BAC}} \text{ OR } (\sin \hat{BAC} =) 0.652024... \quad (A1)$$

$$40.6943...$$

$$\hat{BAC} = 40.7^\circ \quad A1$$

[3 marks]

- (b) **METHOD 1** (cosine rule with $\hat{ABC}$ or $\hat{BAC}$)

attempt to use the cosine rule (M1)

$$24^2 = AB^2 + 17^2 - 2 \cdot 17 \cdot AB \cdot \cos 113^\circ \quad (AB^2 + 13.2848...AB - 287 = 0) \text{ OR}$$

$$17^2 = AB^2 + 24^2 - 2 \cdot 24 \cdot AB \cdot \cos 40.6943...^\circ \quad (AB^2 - 36.3935...AB + 287 = 0) \quad (A1)$$

$$11.5543...$$

$$AB = 11.6 \quad A1$$

METHOD 2 (cosine rule with $\hat{BCA}$)

attempt to use cosine rule (M1)

correct substitution (A1)

$$AB^2 = 17^2 + 24^2 - 2 \cdot 17 \cdot 24 \cdot \cos 26.3056...^\circ \text{ OR } AB^2 = 133.502...$$

$$11.5543...$$

$$AB = 11.6 \quad A1$$

continued...

Question 2 continued

METHOD 3 (sine rule)

attempt to use sine rule

(M1)

correct substitution

(A1)

$$\frac{AB}{\sin 26.3056...^\circ} = \frac{24}{\sin 113^\circ} = \frac{17}{\sin 40.6943...^\circ} \text{ OR } AB = \frac{24 \cdot \sin(180^\circ - 113^\circ - 40.6943...^\circ)}{\sin 113^\circ}$$

11.5543...

AB = 11.6

A1

[3 marks]

Total [6 marks]

3. (a) $(0.708519..., 0.639580...)$

$(0.709, 0.640)$ ($x = 0.709$, $y = 0.640$)

A1A1

[2 marks]

(b) 1.09885...

$x = 1.10$ accept $(1.10, 0)$

A1

[1 mark]

(c) **METHOD 1**

$$\int_0^2 |f(x)| dx$$

(A1)

4.61117...

area = 4.61

A2

METHOD 2

$$-\int_{1.09885...}^2 f(x) dx \text{ OR } \int_{1.09885...}^2 |f(x)| dx \text{ OR } 4.17527...$$

(A1)

$$\int_0^{1.09885...} f(x) dx - \int_{1.09885...}^2 f(x) dx \text{ OR } 0.435901... + 4.17527...$$

(A1)

4.61117...

area = 4.61

A1

[3 marks]

Total [6 marks]

4. $86.4 = 50r^3$ (A1)

$r = 1.2 \left(= \sqrt[3]{\frac{86.4}{50}} \right)$ seen anywhere (A1)

$\frac{50(1.2^n - 1)}{0.2} > 33500$ OR $250(1.2^n - 1) = 33500$ (A1)

attempt to solve their geometric S_n inequality or equation (M1)

sketch OR $n > 26.9045$, $n = 26.9$ OR $S_{26} = 28368.8$ OR $S_{27} = 34092.6$ OR algebraic manipulation involving logarithms

$n = 27$ accept $n \geq 27$ A1

Total [5 marks]

5. recognition that initial population is 15000 (seen anywhere) (A1)

$$P(0) = 15000 \text{ OR } 0.11 \times 15000 \text{ OR } 0.89 \times 15000$$

population after 11% decrease is $15000 \times 0.89 (=13350)$ (A1)

recognizing that $t = 8$ on 1 January 2022 (seen anywhere) (A1)

substitution of their value of t for 1 January 2022 and their value of $P(8)$ into the model (M1)

$$15000 \times 0.89 = 15000e^{8k} \text{ OR } 13350 = 15000e^{8k}$$

$$k = \frac{\ln 0.89}{8} (-0.014566) \text{ (A1)}$$

substitution of $t = 2041 - 2014 (= 27)$ and their value for k into the model (M1)

$$P(27) = 15000e^{-0.0145... \times 27}$$

$$10122.3...$$

$$P(27) = 10100 \text{ (10122)} \text{ (A1)}$$

Total [7 marks]

6.

Note: Do not award any marks if there is clear evidence of adding instead of multiplying, for example ${}^9C_r + (ax)^{9-r} + (1)^r$.

valid approach for expansion (must be the product of a binomial coefficient with $n = 9$ and a power of ax)

(M1)

$${}^9C_r(ax)^{9-r}(1)^r \text{ OR } {}^9C_{9-r}(ax)^r(1)^{9-r} \text{ OR } {}^9C_0(ax)^0(1)^9 + {}^9C_1(ax)^1(1)^8 + \dots$$

recognizing that the term in x^6 is needed

(M1)

$$\frac{\text{Term in } x^6}{21x^2} = kx^4 \text{ OR } r = 6 \text{ OR } r = 3 \text{ OR } 9 - r = 6$$

correct term or coefficient in binomial expansion (seen anywhere)

(A1)

$${}^9C_6(ax)^6(1)^3 \text{ OR } {}^9C_3 a^6 x^6 \text{ OR } 84(a^6 x^6)(1) \text{ OR } 84a^6$$

EITHER

correct term in x^4 or coefficient (may be seen in equation)

(A1)

$$\frac{{}^9C_6}{21} a^6 x^4 \text{ OR } 4a^6 x^4 \text{ OR } 4a^6$$

Set their term in x^4 or coefficient of x^4 equal to $\frac{8}{7}a^5 x^4$ or $\frac{8}{7}a^5$ (do not accept other powers of x)

(M1)

$$\frac{{}^9C_3}{21} a^6 x^4 = \frac{8}{7} a^5 x^4 \text{ OR } 4a^6 = \frac{8}{7} a^5$$

continued...

Question 6 continued

OR

correct term in x^6 or coefficient of x^6 (may be seen in equation) **(A1)**

$$84a^6x^6 \text{ OR } 84a^6$$

set their term in x^6 or coefficient of x^6 equal to $24a^5x^6$ or $24a^5$ (do not accept other powers of x) **(M1)**

$$84a^6x^6 = 24a^5x^6 \text{ OR } 84a = 24$$

THEN

$$a = \frac{2}{7} \approx 0.286(0.285714...) \quad \textbf{A1}$$

Note: Award **A0** for the final mark for $a = \frac{2}{7}$ and $a = 0$.

Total [6 marks]

8. (a) (i) $x = -4$

A1

(ii) attempt to substitute into $y = \frac{a}{c}$ OR table with large values of x OR sketch

of f showing asymptotic behaviour

(M1)

$y = 4$

A1

[3 marks]

(b) (i) $y = \frac{4x+1}{x+4}$

attempt to interchange x and y (seen anywhere)

M1

$xy + 4y = 4x + 1$ OR $xy + 4x = 4y + 1$

(A1)

$xy - 4x = 1 - 4y$ OR $xy - 4y = 1 - 4x$

(A1)

$f^{-1}(x) = \frac{1-4x}{x-4}$ (accept $y = \frac{1-4x}{x-4}$)

A1

continued...

Question 8 continued

(ii) reflection in y -axis given by $f(-x)$ **(M1)**

$$f(-x) = \frac{-4x+1}{-x+4}$$
 (A1)

reflection of their $f(-x)$ in x -axis given by $-f(-x)$ accept “now $-f(x)$ ” **M1**

$$\begin{aligned} (-f(-x)) &= -\frac{-4x+1}{-x+4} \\ &= \frac{-4x+1}{x-4} \text{ OR } \frac{4x-1}{-x+4} \end{aligned}$$
 A1

$$= \frac{1-4x}{x-4} \quad (= f^{-1}(x))$$
 AG

Note: If the candidate attempts to show the result using a particular coordinate on the graph of f rather than a general coordinate on the graph of f , where appropriate, award marks as follows:

MOA0 for eg $(2,3) \rightarrow (-2,3)$

MOA0 for $(-2,3) \rightarrow (-2,-3)$

[8 marks]

continued...

Question 8 continued

- (c) (i) attempt to solve $f(x) = f^{-1}(x)$ using graph or algebraically **(M1)**

$p = -1$ AND $q = 1$ **A1**

Note: Award **(M1)A0** if only one correct value seen.

- (ii) attempt to set up an integral to find area between f and f^{-1} **(M1)**

$$\int_{-1}^1 \left(\frac{4x+1}{x+4} - \frac{1-4x}{x-4} \right) dx \quad \text{ **(A1)** }$$

$= 0.675231\dots$

$= 0.675$ **A1**

[5 marks]

Total [16 marks]

8.

Note: In parts (a) and (b) of this question, candidates may consider either triangle AOD or triangle AOE and work correctly to obtain the answer. Side AD is interchangeable with side AE in the following MS.

- (a) attempt to use right angled trigonometry or sine rule to find AE in terms of r and α

(M1)

$$\tan \alpha = \frac{r}{AE} \text{ OR } \frac{AE}{\sin\left(\frac{\pi}{2} - \alpha\right)} = \frac{r}{\sin \alpha}$$

$$AE = \frac{r}{\tan \alpha} \text{ OR } AE = \frac{r \sin\left(\frac{\pi}{2} - \alpha\right)}{\sin \alpha} \text{ OR } AE = \frac{r \cos \alpha}{\sin \alpha}$$

A1

valid approach to find the area of ADOE

(M1)

2x area of triangle AOE OR area of triangle AED + area of triangle OED OR
OE $\times$ AE

$$\text{Area ADOE} = 2\left(\frac{1}{2} \cdot \frac{r}{\tan \alpha} \cdot r\right) \text{ OR } r \times AE$$

A1

$$\text{Area ADOE} = \frac{r^2}{\tan \alpha}$$

AG

[4 marks]

continued...

Question 8 continued

- (b) (i) recognizing that the sum of the angles of a kite is 2π (M1)

$$\hat{DOE} + \hat{OEA} + \hat{EAD} + \hat{ADO} = 2\pi \text{ OR } 2\alpha + 2 \cdot \frac{\pi}{2} + \hat{DOE} = 2\pi$$

$$\hat{DOE} = \pi - 2\alpha$$

A1

Note: Award **M1A0** if candidate uses degrees (i.e.

$$\hat{DOE} + \hat{OEA} + \hat{EAD} + \hat{ADO} = 360^\circ \text{ or } 2\alpha + 2 \cdot \frac{\pi}{2} + \hat{DOE} = 360^\circ) \text{ and obtains}$$

$$\hat{DOE} = 180^\circ - 2\alpha.$$

- (ii) valid approach to find the area of R (M1)

area of kite – area of sector OR 2(area of triangle AOE – 0.5 area of sector OED)

$$\text{Area of sector} = \frac{1}{2}r^2 \cdot \hat{DOE} \left(= \frac{1}{2}r^2 (\pi - 2\alpha) \right) \text{ seen anywhere} \quad \text{(A1)}$$

$$\text{Area of R} = \frac{r^2}{\tan \alpha} - \frac{1}{2}r^2 (\pi - 2\alpha)$$

A1

Note: Accept $\frac{r^2}{\tan \alpha} - \frac{1}{2}r^2 \cdot \hat{DOE}.$

[5 marks]

continued...

Question 8 continued

- (c) equating their area formula to πr^2 **(M1)**

$$\frac{r^2}{\tan \alpha} - \frac{1}{2}r^2(\pi - 2\alpha) = \pi r^2$$

correct equation in terms of α **A1**

$$\frac{1}{\tan \alpha} - \frac{1}{2}(\pi - 2\alpha) = \pi$$

valid approach to solve the equation **(M1)**

$$\alpha = 0.218979\dots$$

$$\alpha = 0.219$$
 A1

[4 marks]

Total [13 marks]

9.

Note: Do not penalize for inclusion or non-inclusion of endpoints for probabilities using a normal distribution. For example, for $P(T < 55 | T > 40)$ accept $P(T \leq 55 | T > 40)$, $P(T \leq 55 | T \geq 40)$, etc.

(a) recognising to find $P(T > 40)$

(M1)

$$P(T > 40) = 0.574136...$$

$$P(T > 40) = 0.574$$

A1

[2 marks]

(b) attempt to multiply four independent probabilities using their $P(T > 40)$ and

$$P(T < 40)$$

(M1)

$$(1-p)^3 \cdot p \text{ OR } (1-0.574136...)^3 \cdot 0.574136... \text{ OR } (0.425863...)^3 \cdot 0.574136...$$

(A1)

$$0.0443430...$$

$$0.0443, 0.0444 \text{ from 3 sf values}$$

A1

[3 marks]

continued...

Question 9 continued

- (c) (i) recognizing conditional probability (M1)

$$P(T < 55 | T > 40)$$

Note: Award (M1) for an expression or description in context. Accept

$P(T > 40 | T < 55)$ but do not accept just $P(A | B)$.

$$\frac{P(40 < T < 55)}{P(T > 40)} \quad (A1)$$

$$\frac{0.461944...}{0.574136...} \quad (A1)$$

$$P(T < 55 | T > 40) = 0.804590...$$

$$= 0.805 \quad A1$$

- (ii) recognizing binomial probability (M1)

$$X \sim B(n, p)$$

$$n = 10 \text{ and } p = 0.804589... \quad (A1)$$

$$0.0242111..., 0.0240188... \text{ using } p = 0.805$$

$$P(X = 5) = 0.0242 \quad A1$$

[7 marks]

continued...

Question 9 continued

(d) Let $P(T < a) = x$

recognition that probabilities sum to 1 (seen anywhere)

(M1)

EITHER

expressing the three regions in one variable

(M1)

$$x + 0.904 + 2x \text{ OR } P(T < a) + 0.904 + 2P(T < a) \text{ OR } \frac{1}{2}P(T > b) + 0.904 + P(T > b)$$

OR x and $2x$ correctly indicated on labelled bell diagram

$$P(T < a) + 0.904 + 2P(T < a) = 1 \text{ OR } \frac{1}{2}P(T > b) + 0.904 + P(T > b) = 1 \text{ (or}$$

equivalent)

(A1)

OR

expressing either $P(T < a)$ or $P(T > b)$ only in terms of $P(a \leq T \leq b)$

(M1)

$$(P(T < a) =) \frac{1}{3}(1 - P(a \leq T \leq b)) \text{ OR } (P(T > b) =) \frac{2}{3} \cdot (1 - P(a \leq T \leq b))$$

$$x = \frac{1}{3}(1 - 0.904) (= 0.032) \text{ OR } P(T > b) = \frac{2}{3}(1 - 0.904) (= 0.064)$$

(A1)

THEN

$$P(T < a) = 0.032$$

$$a = 22.18167...$$

$$a = 22.2 \text{ accept } 22.1$$

A1

[4 marks]

Total [16 marks]